Escape the Streets Traffic Education Game

Team Geralt
Master of Applied IT
Fontys University of Applied Sciences
Eindhoven

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Abstract

Escape the Streets is a serious game that lets players practise decisions in traffic while delivering packages. This study examines whether participants answer more traffic questions correctly after playing and how they experience the game. Participants completed the same traffic questionnaire before and after playing and rated their enjoyment on a five-point scale after playing. We analysed responses from 10 Fontys students who completed the questionnaire before and after playing. The average score increased from 4.7 to 5.2 out of 6, and all 10 participants rated their enjoyment at 4 or 5 out of 5.

1 Introduction

The starting point for this project was a case about making streets in Eindhoven safer. We focused on traffic education for children from 6-12 years old. We developed Escape the Streets to let players experience traffic situations from three different categories, a pedestrian, cyclist and van driver. Per category there are multiple levels developed.

The player collects a package at point B and has to deliver it to point C within five minutes. During the delivery the player has to take care of crossings, traffic lights and other road users. If you cause a violation it costs a star and triggers a message explaining the mistake. We want to find out whether players understand these situations better after playing and whether they find the game fun. We also wanted to explore engagement.

2 Traffic education and baseline

Dutch primary schools decide how much time to spend on traffic education. Children also learn traffic during everyday trips with their parents or carers [1]. Veilig Verkeer Nederland (VVN) combines classroom activities, practical lessons and home exercises. The materials include digital exercises with feedback [2].

SWOV reports that some primary-school traffic education programmes improve knowledge and show small changes in behaviour, while the evidence varies between programmes [1]. The WEVER evaluation also found that its school data could not establish the effects of individual education programmes [3].

One study close to our game evaluated WegWijsVR. After six weeks of additional virtual practice, the percentage of tasks completed safely in the virtual assessment increased from 16% to 49%. The group continuing the usual school lessons showed a change of approximately one percentage point.. The

children who rated the training gave it an average of 8 out of 10 [4]. These findings give us a reference for virtual traffic training. The different tasks and amount of practice limit direct comparison with our playtest.

For our own study, each participant's score on the form before playing is their starting baseline. We compare it with their score after playing.

3 Research question and hypotheses

Research question. How did participants's scores on traffic questions change after playing Escape the Streets and how much did they enjoy the game?

H1. On average, participants will answer more traffic questions correctly after playing the game compared to before playing the game.

H2. Most participants will report that they enjoyed playing the game.

For H2, we count ratings of 4 or 5 out of 5 as positive. We expected more than half of the participants to give a positive rating.

4 Methodology

4.1 Design and participants

We used a single group with a form before and after playing. The intended audience of *Escape the Streets* is children aged 6 to 12. For this first test, we recruited a mixed group of students from Fontys. Their responses can help us find problems with the game and explore changes in scores before and after playing. Eleven Fontys students completed both forms. We excluded one participant's two submissions because their test was reported to have gone wrong. This left 10 participants in the analysis.

4.2 Questionnaire design

We used Microsoft Forms to create a questionnaire. Participants fill in whether they are completing the form before or after playing. The quiz includes six multiple-choice questions. We chose the cyclist perspective to test with the participants. Each question had five answer options. The images show the situations from the game per question. Correct answers are hidden from participants. When preparing the traffic questions, we looked at the VVN refresher quiz for examples of how to present traffic situations and answer options [5]. We included questions of different difficulty, with some more challenging questions to leave room for improvement before and after playing the game. If participants already score 100% before playing, their scores cannot improve. This is called a ceiling effect.

We used the same traffic questions before and after playing, with the answer options shown in a counterbalanced order. This makes it harder to repeat a choice by remembering where it appeared. Participants can still recognise the question and remember the wording of the answer they chose.

4.3 Procedure

Before playing, participants enter their name and answer the traffic questions. They then receive a short explanation of the controls and play levels 3, 4, 6, and 8 of the cyclist's perspective. They get up to five minutes per level. We selected these levels because these traffic situations most closely match the questions covered at the questionnaire.

After playing participants fill in the same identity as before and answer the same traffic questions and rate how fun the game was. We use their identity to match the two forms. If an email address is available on both submissions, we can use it to check the match.

4.4 Measures and analysis

For each participant who completed both forms, we count the correct answers before and after playing. The difference between the two scores shows how much that participant's score changed. We report the number of matched pairs, the average score before and after play, and how many scores went up, stayed the same or went down. We also report how many participants reached the ceiling effect.

We use the five-point rating for the question: How fun was your experience playing the game? We analyse ratings from submissions marked as after playing and report how many participants chose each option, including the proportion who chose one of the two highest options. We treat this as a measure of enjoyment.

5 Results

5.1 Participants and Questionnaire Responses

The Microsoft Forms questionnaire collected 22 submissions in total: 11 before playing and 11 after playing Escape the Streets [6]. We matched the submissions into 11 before-and-after pairs. One participant's test was reported to have gone wrong, so we excluded both of that participant's submissions.

The names provided by all 11 participants allowed submissions to be matched before playing and after playing the game. With this, we could easily analyse the results of before and after of each participant.

The final sample consisted of 10 participants with complete submissions, 20 in total. The same 10 participants were used for the enjoyment measuring.

5.2 Traffic Knowledge Before and After Playing

Participants answered six multiple-choice questions about cycling and traffic rules before and after playing the game. The maximum possible score was six correct answers.

Over the 10 participants, the average score increased from 4.7 out of 6 before playing (78.3%) to 5.2 out of 6 after playing (86.7%). This was an increase of 0.5 correct answers per participant and 8.3% (Table 1).

Table 1: Questionnaire scores of the 10 participants.

Measure	**Before**	**After**
Mean correct answers (out of 6)	4.7	5.2
Mean percentage correct	78.3%	86.7%
Total correct answers (out of 60)	47	52
Participants with 6/6 correct	5	5

Table 2: Correct answers per question of the 10 participants.

Question topic	**Before**	**After**
1. Yellow traffic light	7/10	8/10
2. Shark teeth and give-way sign	7/10	8/10
3. Priority road and waiting car	8/10	8/10
4. Pedestrian at zebra crossing	8/10	9/10
5. Road position without a cycle path	9/10	10/10
6. When bicycle lights are required	8/10	9/10

5.3 Game Enjoyment and Participant Feedback

All 10 participants rated their enjoyment of the game on a scale from 1 (least fun) to 5 (most fun). Seven participants rated the game 5 out of 5 (70%), and three rated it 4 out of 5 (30%). 1, 2 and 3 were not selected. The mean enjoyment score was 4.7 out of 5, meaning that all 10 participants (100%) selected one of the two highest ratings, 4 and 5. These results indicate that the participants generally enjoyed the game.

Suggestions included adjusting bicycle speed, and making some cycle routes and crossings clearer. Some respondents commented on the strictness or sensitivity of penalties and collisions. One participant suggested that bicycle lights should reset after losing a life, encouraging the player to remember to switch them on again.

6 Discussion and limitations

We tested with Fontys students, while the game is intended for children from 6-12 years old. The students may have different traffic knowledge, reading skills and experience with games. Their feedback can guide changes to the prototype. Testing with children is needed to assess whether the game and questionnaire are suitable for the intended age group.

We used the VVN quiz as a reference when writing the questions. The resulting questionnaire has not been formally validated. The more difficult questions were included to reduce the chance of a ceiling effect, although some participants may still score full marks before playing.

Changing the order of the answer options reduces reliance on remembering an option's position. Participants may still recognise the questions and recall their previous answers. With one group, it is difficult to separate changes due to playing from changes due to seeing the questions twice. Given the time available, we used the same items for this first test. A follow-up study could use two larger sets of validated questions that cover the same topics at a similar difficulty, with one set before play and the other afterwards.

The fun rating measures participants' enjoyment. The form has no separate question on engagement, such as whether the game held their attention or whether they wanted to keep playing. We can use any observations recorded during play as exploratory evidence of engagement.

The average score increased by 0.5 points out of 6, in line with H1. Three participants improved their score, four scored the same and three scored lower. Five participants already scored 100 percent score before playing, so their scores had no room to improve. With all 11 participants included the total number of correct answers went from 53 to 52. For H2 all 10 participants in the analysis chose one of the two highest fun ratings, 4 or 5.

The average increase of 0.5 points out of 6 was small and depended on excluding one participant. This gives us limited evidence that playing the game improved traffic knowledge. In a follow-up study, we would recruit more children aged 6 to 12 and use two sets of validated questions described above.

7 Conclusion

Of the 10 participants included in the analysis, the average score increased from 4.7 to 5.2 out of 6, and the average fun rating was 4.7 out of 5. The average increase of 0.5 points out of 6 was small, and scores decreased slightly when all 11 participants were included. These results give limited evidence that playing the game improved traffic knowledge.

In the next version, we would make the cycle routes and crossings clearer. We would also develop and validate a larger set of traffic questions and test the game with more participants aged 6 to 12. Using different questions of similar difficulty before and after playing would help us see whether children can apply what they practised to new traffic situations.

References

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